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SHORT-PAPER

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OMAR-RQ: Open Music Audio Representation Model Trained with Multi-Feature Masked Token Prediction

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Abstract

Open-source foundation models are essential for advancing music audio understanding and ensuring access to general-purpose representations for music information retrieval. To this end, we present OMAR-RQ, a model trained with self-supervision via masked token prediction using a large-scale dataset with over 330,000 hours of music audio. We experiment with various input features and quantization options, outperforming existing open self-supervised models in music tagging, pitch estimation, chord recognition, beat tracking, segmentation, and difficulty estimation. Finally, we release our training and evaluation pipelines and model weights at <https://github.com/mtg/omar-rq>.

CCS Concepts

• Information systems → Music retrieval.

Keywords

self-supervised learning, music representation learning

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1 Introduction

Currently, there is strong interest in building representation models that map raw audio to compact vectors capturing musical meaning and structure [2, 19, 22, 23]. These representations can be applied to a variety of music analysis tasks such as semantic tagging, structure analysis, and retrieval. Researchers commonly fine-tune these models or utilize their embeddings to solve downstream tasks. This

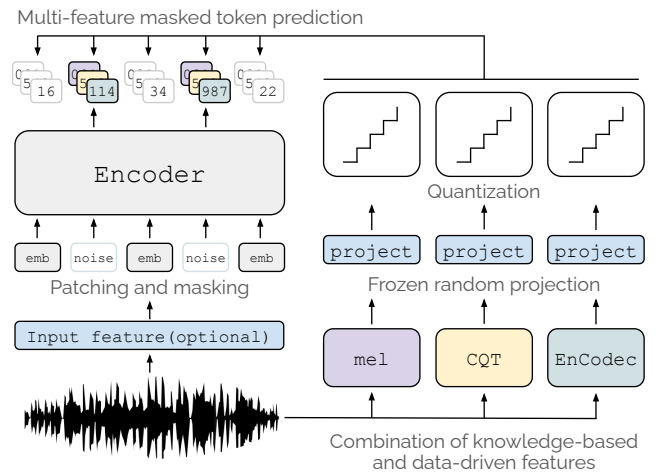


Figure 1: Diagram of the OMAR-RQ architecture.

approach leverages large-scale pre-training to reduce data and compute requirements while enhancing generalizability. We argue that access to a strong, open representation is essential for researchers and practitioners to overcome technical barriers and develop efficient music analysis solutions and applications.

Current models learn representations directly from large-scale data using self-supervised learning (SSL) methods. Popular techniques include contrastive learning [19] or masked prediction frameworks targeting continuous [11] or discrete objectives derived from clustering [23] or random projections [22]. Although recent systems such as [22] report strong performance in downstream tasks using these techniques, they lack open weights, which hinders reproducibility and slows the progress of music analysis research.

In this work, we present OMAR-RQ, an SSL model trained with masked token classification building upon BestRQ [5], exploring various input and target features and quantization settings. We evaluate tokenization strategies applied to waveform, mel spectrograms, constant-Q transforms (CQT), neural encodings [7], and their combinations. Additionally, we experiment with random codebooks and finite scalar quantization [16] settings to optimize performance. We train several OMAR-RQ variations and benchmark them in



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Num.	Random codebook		Pre-training valid. acc.	Downstream task Tagging (mAP)
	Codewords	Usage (%)		
1	32768	75.05%	.257	.461
4	8192	96.46%	.281	.468
16	2048	99.82%	.339	.465

Table 1: Pre-training validation accuracy and downstream music tagging mAP for different codebook settings.

multiple music analysis tasks. Our results show that the proposed models perform competitively with, or outperform, current open masked token prediction systems.

To promote open-source music audio representation models, we release all pre-trained model weights and source code for training and evaluation tools as part of an open research framework under the CC BY-NC-SA 4.0¹ and AGPL-3.0² licenses.

2 Pre-training methodology

Prediction of partially masked input signals is an established SSL paradigm in audio [11]. Inspired by the NLP literature, some approaches “tokenize” audio by quantizing handcrafted or learned audio features to approach the reconstruction as a classification task. Established quantization methods include k-means clustering [23] or random codebook projection [5]. We continue this research line by exploring various audio tokenization configurations.

BEST-RQ. This method operates by projecting a sequence of audio input features (e.g., mel spectrograms) into a lower-dimensional space using a dense layer with random non-trainable weights [5]. Then, a codeword is assigned by nearest-neighbor search in a codebook of random vectors drawn from the uniform distribution. The audio encoder is trained to predict the codewords corresponding to masked regions from the rest of the input signal.

Multi-codebook. Previous works relying on audio quantization reported suboptimal performance associated with codebook underutilization [12]. In our work, we propose utilizing multiple codebooks with different initializations. This transforms our pre-training task into a multi-label classification problem, where each codebook provides a target label.

Multi-feature. Since the goal of the multi-codebook method is to enrich the coverage of the target space, we hypothesize that using different audio input features may further increase the representation power of the model. We propose computing multiple targets by quantizing different input features. In our work, we experiment with mel-spectrograms, CQT, EnCodec features [7], and the raw waveform as input features.

Finite Scalar Quantization (FSQ). We apply FSQ as a drop-in replacement for the random codebook assignments of BEST-RQ. This method scales the low-dimensional projections to a given range by $\lfloor \frac{x}{2} \rfloor \cdot \tanh(x)$, and quantizes it via the round operator [16].

3 Experimental setup

In this section, we introduce the details related to model training.

Features		Downstream tasks	
Input	Target	Tagging (mAP)	Pitch (acc.)
mel	mel	.469	.824
cqt	cqt	.419	.852
enc	enc	.442	.878

Table 2: Downstream performance for OMAR-RQ trained with mel-spectrograms, CQT, or EnCodec embeddings.

3.1 Pre-training dataset

We pre-train our models in 330,000 hours of music audio extracted from 6.6 million YouTube videos. Following previous work [1, 2], we relied on the Discogs³ music database to create our dataset. Discogs releases monthly editorial metadata dumps under a CC0-BY license, including YouTube URLs. This process favors building a corpus of high-quality and diverse music audio.⁴

3.2 Implementation details

All of our models use the Conformer architecture [10] considering two configurations: a *base* model with 12 layers, 512 embedding dimensions, and 90 million parameters, and a *large* model with 24 layers, 1024 embedding dimensions, and 580 million parameters. The *base* architecture is used in our development experiments, and the *large* architecture is used in our final models reported in Table 4. Both architectures use ROPE positional encoding [20], flash attention [6], and automatic mixed-precision training.⁵ Additionally, we use DeepNorm ($\alpha = 2.632$ and $\beta = 0.022$) to prevent training collapse observed in the *large* architectures [21]. All the models are trained for 400,000 steps, using cosine annealing with 30,000 warm-up steps and a maximum learning rate of 1×10^{-4} . We use dropout and weight decay values of 0.2 and 1×10^{-2} for regularization. Our models are trained with a combined batch size of 256 30-second samples using 4 Nvidia 64GB H100 GPUs for the *base* version and 8 for the *large* one. Due to the GPU memory constraints, the high-rate variants used a combined batch size of 160. Training takes over 3 days for our *base* models and 5 days for the *large* ones.

Our BEST-RQ configuration follows previous works [5, 22]. Our models use 30-second segments from which 60% is masked in chunks of 0.4 seconds. We experimented with Gaussian noise and waveform shuffling as our masking strategies, obtaining better

³<https://www.discogs.com/>

⁴The Discogs database has 15 music genres categorized into 700 styles. Our repository provides a complete list of the YouTube URLs used for pre-training.

⁵We used PyTorch Lightning’s bf16-mixed precision.

Input	Target	Tagging	Pitch
enc	enc	.411	.878
enc	enc/mel	.445	.898
enc	enc/cqt	.433	.892
enc	enc/cqt/audio	.433	.889
enc	enc/mel/audio	.444	.895
enc	enc/mel/cqt	.446	.900
enc	enc/mel/cqt/audio	.446	.878

Table 3: Comparison of target features in the downstream tasks of music tagging and pitch estimation.

¹<https://creativecommons.org/licenses/by-nc-sa/4.0/>

²<https://www.gnu.org/licenses/agpl-3.0.en.html>

Model	Size M	RF s	Input	Target	Quantizer	Rate Hz	Tagging <i>mAP</i> ↑	Difficulty <i>MSE</i> ↓	Pitch <i>acc.</i> ↑	Chord <i>acc.</i> ↑	Beat <i>F1</i> ↑	Structure <i>acc.</i> ↑
MAEST	86	30	mel	genre labels	N/A	3.9	.491	-	-	-	-	-
MERT	330	5	audio	cqt, enc	k-means	75	.441	1.89	.922	.609	.868	.393
MusicFM FMA	330	30	audio	mel	RQ	25	.448	1.78	.857	.644	.847	.635
base	580	30	mel	mel	RQ	15.625	.482	1.65	.892	.657	.783	.647
multi-codebook	580	30	mel	mel	4 × RQ	15.625	.488	1.66	.897	.675	.775	.639
multi-feature	580	30	audio	enc/mel/cqt/audio	4 × RQ	18.75	.467	1.76	.938	.734	.833	.623
high rate	580	30	audio	enc/mel/cqt/audio	4 × RQ	25	.463	1.79	.932	.728	.848	.628
high rate FSQ	580	30	audio	enc/mel/cqt/audio	4 × FSQ	25	.463	1.71	.940	.749	.855	.628

Table 4: Left: model details, bold identifies OMAR-RQ variants. Right: Downstream metrics, bold indicates the best result.

results with the former method. We use a single codebook with 8192 codewords of 16 dimensions in our *base* configuration, and four codebooks with the same parameters in the **multi-codebook**, **multi-feature**, and **high rate** versions. For the **high rate FSQ** version, we report the performance with 5 channels and 6 levels.

4 Evaluation

To evaluate the representations learned by the proposed and existing open SOTA models, we consider several downstream tasks covering both track-level and time-based musical notions.

4.1 Downstream protocol and tasks

Following common practice in the literature, we use two-layer MLPs with 512 hidden units as probes to evaluate the proposed representations. For track-level tasks, the embeddings extracted from the different models are averaged over the time dimension to obtain one vector representation per track. For time-based tasks, the MLP operates on timestamps independently, and optional post-processing is applied following the standard practice for each task.⁶

Music tagging is a multi-label classification task where each track is assigned labels related to musical genre, mood, era, or instrumentation. We use the MagnaTagATune dataset containing 25,887 tracks with tag annotations [13], using an established split with tracks annotated by the 50 most frequent tags [22], and report results in terms of macro-averaged mean Average Precision (mAP).

Pitch estimation consists of estimating the fundamental frequency of musical signals. We use the NSynth dataset containing 305,979 4-second single-note audio clips ranging from C-2 to B-7 (128 pitch classes) [8]. The results are reported in terms of accuracy.

Chord recognition is the harmony analysis task of assigning chord labels to successive audio frames. We use four standard chord annotation datasets, Isophonics [15], Billboard [4], USPOP2002 [3], and RWC-Pop [9]. The data is split into 60% training, 20% validation, and 20% test sets, using 30-second audio segments extracted with 50% overlap. The model predicts frame-level probabilities over 25 classes: 12 major, 12 minor, and a ‘none’ label. Evaluation is based on major/minor weighted accuracy using *mir_eval*.

Beat tracking estimates the temporal locations of beats in a musical signal. We use the Harmonix Set dataset [17] with 912 full tracks for training and validation, and the GTZAN dataset [14] with 1,000 30-second excerpts for testing. We train and validate in 30-second segments extracted from the full tracks with a 50%

overlap. For testing, we directly report the results on the GTZAN excerpts. We used the F-measure metric as in previous work [22].

Structure segmentation consists of dividing a music recording into non-overlapping sections and assigning each one a label such as ‘verse’ or ‘chorus’. Following [22], we use two classifiers: one for frame-level functional classes and one for boundaries, with 200 ms frames. We divide the Harmonix Set [17] into 60% training, 20% validation, and 20% test. Table 4 shows the frame-wise accuracy. Unlike [22], we do not use data augmentation or CTC loss. We release the split for fair comparisons.

Piano difficulty estimation is a regression task in which each music piece is assigned a difficulty score. We use the PianoSyllabus Dataset [18] containing graded annotations based on piano curricula. The probe is trained in the canonical split using an ordinal regression loss and reports mean squared error (MSE) following [18].

5 Results

In this section, we discuss our experimental results.

Multi-codebook settings. Table 1 compares three *base* models targeting the same number of codewords arranged in 1, 4, and 16 parallel codebooks. We observe that a larger codebook allows for more detailed targets, but leads to codebook underutilization. On the other hand, many parallel codebooks allow for a more efficient use of the codebook at the expense of less accurate quantization.

Comparison of features. We assess the performance of different features both as input and training targets. Particularly, we train *base* architectures using mel-spectrograms, CQT, and EnCodec embeddings before quantization [7]. Table 2 compares the models in music tagging and pitch estimation. Interestingly, the EnCodec embeddings perform higher than the CQT, which explicitly contains pitch class information. Upon further inspection, we observed that the CQT tokenizer frequently assigned the same codeword to samples from the same instrument and close pitches in the higher octaves, resulting in poorer high-frequency performance.

Multi-feature targets. Table 3 compares multiple *base* architectures trained with a fixed input and a systematic combination of features as target. Our results show that using multiple features as targets improves performance for all the cases except for the waveform, which decreases the performance in pitch estimation.

Benchmark with SOTA models. We present a benchmark comparing variants of OMAR-RQ with relevant SOTA models. We consider MusicFM [22], which also follows a BEST-RQ approach and is available in one variant with open weights. Also, we considered MERT [23], combining tokens from CQT clusters and EnCodec.

⁶Our repository provides the hyperparameters used for each task.

Finally, we consider MAEST, an audio transformer trained on the same dataset as OMAR-RQ and targeting music style tags from Discogs suitable for tagging tasks [2]. The main characteristics and downstream metrics for the models are summarized in Table 4. **OMAR-RQ multi-codebook** obtains the best performance in music tagging among the open SSL models, although MAEST performs better thanks to its supervised training and task alignment with the Discogs genre tags. For pitch estimation and chord recognition, **OMAR-RQ high-rate** achieves the highest performance among the compared models, suggesting that the additional temporal resolution and more musically-aware frequency representation benefit these tasks. For beat tracking, we observe that the results are favoured by a higher temporal resolution, resulting in MERT performing the best, followed by the high-rate variants of our models. For structure segmentation, we observe small variations among our models. Interestingly, variants with mel-spectrogram targets perform better for this task. In the current probe evaluation setup, the structure recognition task tends to perform better with the same representations as in music tagging instead of beat-tracking (which is also a time-based task). This suggests that representations focused on temporal details are not as useful as those that perform well on the classification of semantic categories for this task. Lastly, the *base* model achieved the lowest MSE value in difficulty estimation, following the same trend as in music tagging.

Overall, we observe two distinct patterns in the downstream evaluation: models with lower feature rate only modeling timbre information (mel) perform better on high-level music understanding tasks (tagging, difficulty, and structure). In contrast, models with a higher feature rate that also model audio detail (audio, enc) and tonal (cqt) information perform better in low-level tasks (pitch, chord, and beat tracking). Finally, OMAR-RQ achieves the highest performance among the considered open models, although other works have reported higher results in undisclosed settings.

6 Conclusion

In this work, we build upon the BEST-RQ paradigm to create a new generation of open-source SSL music audio representation models for music. These models can be applied to various music analysis tasks, and we envision their usage by music information retrieval researchers and practitioners and beyond. We expect them to foster further advancements in representation learning research and downstream music audio analysis tasks, with potential for developing final applications. To develop our models, we experimented with SOTA SSL methodologies based on masked token prediction to build models with a higher overall performance across a range of benchmark music audio analysis tasks. Our technical contributions include experimental results utilizing parallel codebooks, multi-feature targets, and novel quantization methods.

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References

- [1] Pablo Alonso-Jiménez, Dmitry Bogdanov, and Xavier Serra. 2022. Music Representation Learning Based on Editorial Metadata from Discogs. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [2] Pablo Alonso-Jiménez, Xavier Serra, and Dmitry Bogdanov. 2023. Efficient Supervised Training of Audio Transformers for Music Representation Learning. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [3] Adam Berenzweig, Beth Logan, Daniel PW Ellis, and Brian Whitman. 2004. A Large-scale Evaluation of Acoustic and Subjective Music-similarity Measures. *Computer Music Journal* (2004), 63–76.
- [4] John Ashley Burgoyne, Jonathan Wild, and Ichiro Fujinaga. 2011. An Expert Ground Truth Set for Audio Chord Recognition and Music Analysis. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [5] Chung-Cheng Chiu, James Qin, Yu Zhang, Jiahui Yu, and Yonghui Wu. 2022. Self-Supervised Learning with Random-Projection Quantizer for Speech Recognition. In *Int. Conf. on Machine Learning (ICML)*.
- [6] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. 2022. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [7] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. 2022. High Fidelity Neural Audio Compression. *arXiv preprint arXiv:2210.13438* (2022).
- [8] Jesse Engel, Cinjon Resnick, Adam Roberts, Sander Dieleman, Mohammad Norouzi, Douglas Eck, and Karen Simonyan. 2017. Neural Audio Synthesis of Musical Notes with Wavenet Autoencoders. In *Int. Conf. on Machine Learning (ICML)*.
- [9] Masataka Goto, Hiroki Hashiguchi, Takuichi Nishimura, and Ryuichi Oka. 2002. RWC Music Database: Popular, Classical and Jazz Music Databases. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [10] Anmol Gulati, James Qin, Chung-Cheng Chiu, Niki Parmar, Yu Zhang, Jiahui Yu, Wei Han, Shibo Wang, Zhengdong Zhang, Yonghui Wu, et al. 2020. Conformer: Convolution-Augmented Transformer for Speech Recognition. *arXiv preprint arXiv:2005.08100* (2020).
- [11] Po-Yao Huang, Hu Xu, Juncheng Li, Alexei Baevski, Michael Auli, Wojciech Galuba, Florian Metze, and Christoph Feichtenhofer. 2022. Masked Autoencoders that Listen. *Advances in Neural Information Processing Systems* 35 (2022).
- [12] Adrian Łańcucki, Jan Chorowski, Guillaume Sanchez, Ricard Marxer, Nanxin Chen, Hans JGA Dolfing, Sameer Khurana, Tanel Alumäe, and Antoine Laurent. 2020. Robust Training of Vector Quantized Bottleneck Models. In *2020 Int. Joint Conf. on Neural Networks (IJCNN)*.
- [13] Edith Law, Kris West, Michael I Mandel, Mert Bay, and J Stephen Downie. 2009. Evaluation of Algorithms Using Games: The Case of Music Tagging.. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [14] Ugo Marchand, Quentin Fresnel, and Geoffroy Peeters. 2015. GTZAN-rhythm: Extending the GTZAN Test-Set With Beat, Downbeat and Swing Annotations. (2015).
- [15] Matthias Mauch, Chris Cannam, Matthew Davies, Simon Dixon, Christopher Harte, Sefki Kolozali, Dan Tidhar, and Mark Sandler. 2009. OMRAS2 Metadata Project 2009. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [16] Fabian Mentzer, David Minnen, Eirikur Agustsson, and Michael Tschannen. 2024. Finite Scalar Quantization: VQ-VAE Made Simple. In *Int. Conf. on Learning Representations (ICLR)*.
- [17] Oriol Nieto, Matthew C McCallum, Matthew EP Davies, Andrew Robertson, Adam M Stark, and Eran Egozy. 2019. The Harmonix Set: Beats, Downbeats, and Functional Segment Annotations of Western Popular Music.. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [18] Pedro Ramoneda, Minhee Lee, Dasaem Jeong, Jose J. Valero-Mas, and Xavier Serra. 2025. Can Audio Reveal Music Performance Difficulty? Insights From the Piano Syllabus Dataset. *IEEE Transactions on Audio, Speech and Language Processing* 33 (2025), 1129–1141.
- [19] Janne Spijkervet and John Ashley Burgoyne. 2021. Contrastive Learning of Musical Representations. In *Int. Society for Music Information Retrieval Conf. (ISMIR)*.
- [20] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. Roformer: Enhanced Transformer with Rotary Position Embedding. *Neurocomputing* 568 (2024), 127063.
- [21] Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Dongdong Zhang, and Furu Wei. 2024. Deepnet: Scaling Transformers to 1,000 Layers. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).
- [22] Minz Won, Yun-Ning Hung, and Duc Le. 2024. A Foundation Model for Music Informatics. In *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*.
- [23] Li Yizhi, Ruibin Yuan, Ge Zhang, Yinghao Ma, Xingran Chen, Hanzhi Yin, Chenghao Xiao, Chenghua Lin, Anton Ragni, Emmanouil Benetos, et al. 2023. MERT: Acoustic music understanding model with large-scale self-supervised training. In *Int. Conf. on Learning Representations (ICLR)*.